25901192

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSECSC200 - DATA MINING
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Scheme of Evaluation

Part A

Very Short Answer Questions
Answer all questions
Each question carries **2** marks

1. Define decision tree. [U] / [CO1]

- **1 mark → Correct definition (tree-like model for decision making / classification / prediction)**
- **1 mark → Key feature (nodes, branches, splitting based on features)**

2. Define lift in correlation. [U] / [CO2]

- **1 mark → Definition (measure of strength of association between variables/items)**
- **1 mark → Formula or interpretation (lift(A, B) = P(A ∩ B) / (P(A) · P(B)) Lift > 1 indicates positive correlation)**

3. Differentiate between Single, Complete, and Average Linkage Methods. [An] / [CO3]

- **1 mark → Correct explanation of Single, Complete, and Average Linkage Methods**
- **1 mark → Clear differentiation (based on distance: minimum, maximum, average)**

4. Describe methods to handle missing values. [U] / [CO1]

- **1 mark each → Any two valid methods (deletion, mean/median/mode imputation, predict most probable value, interpolation, etc.) with a short description**

5. State the role of clustering in real-world applications. [K] / [CO3]

- **1 mark → Definition/role (grouping similar data points)**
- **1 mark → Any one or two applications (customer segmentation, image processing, fraud detection, etc.)**

Part B

Short Answer Questions
Answer any 5 questions
Each question carries 5 marks

6. Discuss about association analysis. [U] / [CO1]

- **1 mark – Definition**
  → Finds relationships/patterns among items in large datasets

- **2 marks – Key Concepts (any 2–3 points)**
  → Itemset
  → Support (frequency of occurrence)
  → Confidence (strength of rule)
  → Lift (correlation measure)
- **1 mark – Algorithm**
  → Apriori Algorithm explanations
- **1 mark – Example**
  → Market basket analysis (e.g., bread → butter)

7. Differentiate between OLTP and OLAP. [U] / [CO2]

- **1 mark – OLTP Definition**
  → Used for day-to-day transaction processing
- **1 mark – OLAP Definition**
  → Used for data analysis and decision making
- **3 marks – Differences (any 3 points)**

8. Compare SVM and KNN classification methods. [U] / [CO3]

- **1 mark – SVM concept**
  → Finds optimal hyperplane, margin maximization
- **1 mark – KNN concept**
  → Instance-based, classifies using nearest neighbours
- **3 marks – Comparison (any 3–4 points)**

9. Illustrate how Data Mining can be applied in the banking or retail sector. [A] / [CO1]

- **1 mark – Definition/Introduction**
  → Data mining: extracting useful patterns/knowledge from large datasets
- **3 marks – Banking/Retail applications** (1 mark for each relevant point with brief explanation)
- **1 mark – Example**
  →Simple example (e.g., bank fraud alert system or product recommendation)

10. Explain the major issues in Data Mining with suitable examples. [U] / [CO1]

- **2 mark – Definition/Introduction**
  → Data mining: extracting useful patterns/knowledge from large datasets
- **3 marks – Major Issues** (1 mark for each issue with brief explanation)
  → Mining methodology and user interaction issues
  → Performance issues
  → Issues relating to the diversity of database types

11. Explain four widely used measures for distance between Clusters in hierarchical clustering. [U] / [CO3]

- **1 mark –Introduction**
  → Distance measures define how similarity/dissimilarity between clusters is calculated in hierarchical clustering
- **4 marks – Measures of Distance** (1 mark each for any four measures with brief explanation)
  → Single Linkage (Nearest Neighbor)
  → Complete Linkage (Farthest Neighbor)
  → Average Linkage

→ Mean Distance (between Centroids)

12. Explain fact table and dimension table. [U] / [CO2]

- **1 mark –Introduction**
  → Data warehouse tables are categorized into fact tables and dimension tables for analysis.
- **2 marks – Fact Table**
  → 1 mark for definition + 1 mark for features/example
- **2 marks – Dimension Table**
  → 1 mark for definition + 1 mark for features/example

Part C

Essay Questions

Answer any 1 questions

Each question carries 15 marks

13. Explain Association Rule Mining with support and confidence. [U] / [CO2]

- **2 marks – Introduction/Definition**
  → Association rule mining finds relationships/patterns among items in large datasets
- **3 marks – Basic Concepts**
  → Itemset (set of items)
  → Frequent itemset
  → Rule form: X→Y
- **4 marks – Measures (with explanation & formula)**
  → **Support**: frequency of occurrence of itemset
  → **Confidence**: likelihood that Y occurs given X
  → (1–2 marks for formula + explanation)
- **4 marks – Algorithm (brief)**
  → Apriori Algoithm
- **2 marks – Example**
  → Market basket analysis (e.g., bread → butter) with support/confidence

14. Describe the Naïve Bayes Classification algorithm with suitable example. [U] / [CO3]

- **2 marks – Introduction/Definition**
  → Probabilistic classifier based on Bayes' theorem
- **3 marks – Bayes Theorem Explanation**
  → $P(H|X) = P(X|H)P(H)/P(X)$
  → Meaning of prior, likelihood, posterior
- **3 marks – Naïve Assumption**
  → Features are conditionally independent
- **5 marks – Algorithm Steps**
  → Calculate prior probabilities
  → Compute likelihood
  → Apply formula
  → Choose class with highest probability
- **2 marks – Example**
  → Spam detection / simple dataset classification
